Project – Airline Reservation System

Object Oriented Programming

(CSEG1044)

Phase 0: Proposal and Backlog

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1. Project Overview

Project Title: Airline Reservation System

- **Problem Statement:** Managing flight schedules, passenger bookings, and seat allocations across separate files leads to data fragmentation, overbooking, and scheduling conflicts. This Java desktop application centralizes these records into a single structured system to streamline the flight reservation workflow from route creation to final seat assignment.
- **Target Users & Their Capabilities:**
    - **Admin (Airline Manager):** Has full access to the system. Can add/edit Flight details, create new Routes, view all Passenger bookings, update booking statuses (e.g., Pending, Confirmed, Cancelled), and manage Seat Allocations.
    - **Passenger:** Has restricted access. Can view their own profile, browse available Flights, submit a Booking for a flight, and view their booking status and assigned seat.

2. Domain Entities (The Data Model)

- **Passenger:** Stores passengerId, name, email, passportNumber, nationality, dateOfBirth.
- **Route:** Stores routeId, origin, destination, distance, estimatedDuration.

- **Flight:** Stores flightId, routeId (foreign key), airplaneModel, capacity, departureDateTime, arrivalDateTime, ticketPrice.
- **Booking:** Stores bookingId, passengerId, flightId, bookingDate, status (Enum: PENDING, CONFIRMED, CANCELLED).
- **Seat Allocation:** Stores seatId, bookingId, seatNumber, classType (Economy, Business), checkInStatus.

3. Operational Workflow

Primary End-to-End Workflow (The "Happy Path")

1. **Registration:** The Admin logs in and registers a new "Route" in the system.
2. **Flight Creation:** The Admin creates a new "Flight" associated with that Route, specifying the schedule, capacity, and price.
3. **Booking:** A "Passenger" logs into their dashboard, sees the active Flights, and clicks "Book". The system creates a "Booking" record.
4. **Review & Confirm:** The Admin views all bookings for the Flight, reviews capacity, and changes the booking status to "Confirmed".
5. **Scheduling:** The Admin selects the confirmed Booking and creates a "Seat Allocation", assigning a specific seat number. The Passenger can now see this schedule on their dashboard.

4. Scope Definition

- **In-Scope (Version 1):** User authentication (basic login), CRUD operations for all 5 core entities, role-based dashboards (Admin vs. Passenger), booking status tracking, seat scheduling, and basic validations (e.g., preventing a passenger from double-booking the same flight).
- **Out-of-Scope (Version 1):** Live payment gateway integration, real-time flight radar tracking, automated email/SMS ticket delivery, complex dynamic pricing algorithms, and third-party travel agency API integration.

5. Team Organization

To ensure equal distribution of work and hit all course requirements, team members are assigned to the following specific tracks:

Team Set	Core Focus	Specific Deliverables for the Project
Set 1 (Systems Analysts)	Requirements & Design	Draft the final Use Case descriptions, create the Use Case Diagram, design the UML Class Diagram, and maintain the README.md documentation throughout the project.

Team Set	**Core Focus**	**Specific Deliverables for the Project**
Set 2 (Backend Developers)	Domain Model & Services	Code the plain Java objects (POJOs) for the 5 entities. Write the Service classes that handle business logic (e.g., BookingService.java which checks flight capacity before saving).
Set 3 (Database Engineers)	Database & DAO Layer	Design the SQL table schema (DDL). Write the DAO (Data Access Object) classes using JDBC to execute INSERT, SELECT, UPDATE, and DELETE queries. Manage database connection utilities.
Set 4 (Frontend Developers)	Swing UI & Integration	Design the Java Swing GUI (Frames, Panels, Buttons, Tables). Wire the UI buttons to call the methods in the Service layer. Ensure the app is packaged correctly for the final demo.